

Keys

Superkey

- A set of attributes such that …
- If tuples have the same superkey values,
- Then tuples are the same.



Find the superkeys

Student (fname, lname, netid, classification, phone, birthdate)

Find the superkeys

Teaches (instructor, course)

Finding superkeys correctly

Need to consider semantics of schema, not just the data in a single instance.

However, you might be given just the data, and you need to infer the intended semantics.



Are relations guaranteed to have a superkey?

Superkey definition

A set K of attributes is a superkey of R iff

- For every instance R' of R,
- For all tuples t_1 and t_2 in R',
- If $t_1[K] = t_2[K]$, then $t_1 = t_2$.

Key = Candidate key

A minimal superkey – i.e., no subset is a key

Find the keys

Student (fname, lname, netid, classification, phone, birthdate)

Keys in relations vs. tables

Superkey:

- A set of attributes such that …
- If tuples have the same superkey values,
- Then tuples are the same.

Tables allow duplicate tuples. When are tuples “the same”?

The exact same tuple, not just looks the same.

I.e., same physical row.

Primary key

One key chosen from all keys

Significance of primary key

Relations (relational model)

- No different than any other key

Tables (database)

- Used as tuple identifier
- Treated specially in implementation for efficiency

PK in Schema

Product

prod_name	prod_price	prod_manufacturer
Gizmo	$19.99	GizmoWorks
Powergizmo	$39.99	GizmoWorks
Widget	$19.99	WidgetsRUs
HyperWidget	$203.99	Hyper
VaporWare	NULL	NULL

Product (prod_name, prod_price, prod_manufacturer)

Product (prod_name: **string**, prod_price: **money**, prod_manufacturer: **string**)

Keys as constraints

Can designate PK and other keys as DB constraints.

DB enforces property:

If key values are the same, then tuples are the same.